

Bile Acids Analysis Results

[분석 결과 보고서]

Client	HEM pharma
Company Institution	HEM pharma
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Number of Samples	34

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CONTENTS

	Bile acids		00 Sample information
			01 Sample preparation
			02 Instrument information
			03 Metabolite analysis results
	Appendix		01 Sample preparation and Data analysis

00. Sample information

Sample type	Feces
Number of Samples	34
Analysis target	Bile acids
Analytical instrument	LC-MS/MS

» LC-MS/MS : Liquid Chromatography Triple Quadrupole Mass Spectrometry

01. Sample preparation

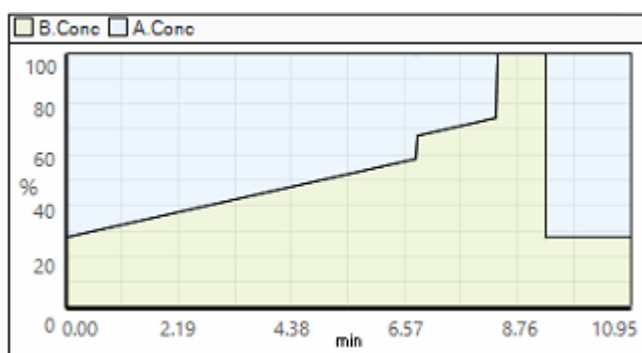
No.	Sample ID	Weight (mg)	50% Methanol (mL)
1	FD-599	48.75	0.975
2	FD-602	49.30	0.986
3	FD-605	48.64	0.973
4	FD-606	49.08	0.982
5	FD-607	49.49	0.990
6	FD-610	49.85	0.997
7	FD-616	48.36	0.967
8	FD-622	49.52	0.990
9	FD-624	48.35	0.967
10	FD-629	48.63	0.973
11	FD-642	49.58	0.992
12	FD-646	48.92	0.978
13	FD-648	48.61	0.972
14	FD-649	49.49	0.990
15	FD-706	49.40	0.988
16	FD-737	49.61	0.992
17	FD-762	48.49	0.970
18	FD-765	48.93	0.979
19	FD-775	48.55	0.971
20	FD-790	49.86	0.997
21	FD-791	49.08	0.982
22	FD-803	49.70	0.994
23	FD-868	48.91	0.978
24	FD-916	48.80	0.976
25	FD-920	50.00	1.000
26	FD-921	48.76	0.975
27	FD-926	48.95	0.979
28	FD-945	48.23	0.965
29	FD-952	49.49	0.990
30	FD-1016	48.23	0.965
31	FD-1033	49.58	0.992
32	FD-1046	49.15	0.983
33	FD-1078	49.12	0.982
34	FD-1079	48.64	0.973

» Weight (mg) : 분석에 사용한 시료의 양

» Methanol (mL) : 분석에 사용한 50% 메탄올의 양

02. Instrument information

Analytical instrument	Shimadzu LCMS-8050
Column information	ACE EXCEL C18-AMIDE Column (1.7 μ m, 2.1 mm X 75 mm)
Run time	10.95 min / sample
Column temperature	45 °C
Mobile phase	(A) Water with 0.01% (v/v) Formic acid (B) Acetonitrile/Methanol 90/10 (v/v)
Flow rate	0.65 mL / min
Injection volume	1 μ L



No.	Time (min)	B.Conc (%)
1	0.00	27.30
2	6.80	58.50
3	6.81	67.20
4	8.35	74.80
5	8.36	100.00
6	9.30	100.00
7	9.31	27.30
8	10.95	27.30

Figure 1. Gradient system used in the LC-MS/MS method

» (Fig. 1) 분석 시간(Run time)에 따른 이동상(Mobile phase) 변화

03. Metabolite analysis results

Sample information		Concentration (μmol/g)															TUDCA+THDCA
No.	Sample ID	CA	CDCA	GCA	GCDCA	TCA	TCDC	DCA	HDCA	LCA	UDCA	GDCA	GLCA	GUDCA	TDCA	TLCA	
1	FD-599	0.008	0.423	0.000	0.001	0.000	0.000	0.432	0.113	0.511	0.078	0.001	0.000	0.000	0.000	0.000	0.000
2	FD-602	0.155	0.452	0.000	0.002	0.000	0.000	0.320	0.113	0.304	0.078	0.000	0.000	0.000	0.000	0.000	0.000
3	FD-605	0.224	1.031	0.002	0.004	0.000	0.000	1.531	0.113	0.785	0.078	0.007	0.000	0.001	0.001	0.000	0.000
4	FD-606	0.000	0.954	0.000	0.001	0.000	0.000	0.869	0.113	1.835	0.078	0.002	0.001	0.000	0.000	0.000	0.000
5	FD-607	0.002	0.598	0.001	0.005	0.000	0.000	0.627	0.113	1.752	0.078	0.003	0.001	0.001	0.000	0.000	0.000
6	FD-610	1.304	0.887	0.005	0.006	0.003	0.002	1.370	0.113	0.821	0.078	0.005	0.000	0.001	0.003	0.000	0.000
7	FD-616	0.208	1.289	0.002	0.003	0.000	0.001	1.176	0.113	0.903	0.078	0.004	0.000	0.001	0.001	0.000	0.000
8	FD-622	0.001	1.416	0.000	0.001	0.000	0.000	0.617	0.113	0.931	0.078	0.002	0.001	0.000	0.000	0.000	0.000
9	FD-624	0.013	1.146	0.005	0.004	0.002	0.001	0.492	0.113	1.269	0.078	0.001	0.000	0.000	0.000	0.000	0.000
10	FD-629	1.825	0.712	0.001	0.002	0.002	0.001	0.995	0.113	0.342	0.078	0.001	0.000	0.000	0.001	0.000	0.000
11	FD-642	1.514	1.232	0.004	0.002	0.007	0.002	2.602	0.113	0.691	0.537	0.005	0.000	0.002	0.010	0.001	0.001
12	FD-646	3.418	1.610	0.006	0.006	0.013	0.007	1.570	0.113	0.231	0.325	0.005	0.000	0.001	0.007	0.000	0.001
13	FD-648	0.000	0.184	0.000	0.000	0.000	0.000	0.082	0.113	0.063	0.078	0.001	0.000	0.000	0.000	0.000	0.000
14	FD-649	0.004	1.121	0.000	0.000	0.000	0.000	2.436	0.113	2.321	0.078	0.003	0.000	0.000	0.000	0.000	0.000
15	FD-706	0.019	9.769	0.001	0.001	0.003	0.001	3.260	0.113	4.688	0.078	0.008	0.002	0.000	0.001	0.000	0.000
16	FD-737	0.008	0.482	0.004	0.007	0.000	0.001	0.418	0.113	0.414	0.078	0.003	0.000	0.000	0.000	0.000	0.000
17	FD-762	0.006	0.493	0.000	0.001	0.000	0.000	1.148	0.113	1.204	0.078	0.003	0.001	0.000	0.002	0.001	0.000
18	FD-765	0.000	0.414	0.000	0.000	0.000	0.000	0.470	0.113	1.022	0.078	0.001	0.001	0.000	0.000	0.000	0.000
19	FD-775	0.003	2.258	0.001	0.002	0.002	0.001	1.061	0.113	1.257	0.078	0.004	0.001	0.000	0.017	0.004	0.000
20	FD-790	0.005	0.057	0.000	0.002	0.000	0.001	0.109	0.113	0.217	0.078	0.000	0.000	0.002	0.000	0.000	0.000
21	FD-791	0.006	1.053	0.001	0.002	0.000	0.000	0.670	0.113	0.612	0.078	0.003	0.000	0.001	0.000	0.000	0.000
22	FD-803	0.000	1.839	0.000	0.001	0.000	0.000	0.502	0.113	1.097	0.078	0.003	0.001	0.000	0.000	0.000	0.000
23	FD-868	0.002	1.066	0.000	0.000	0.000	0.000	0.780	0.113	0.572	0.078	0.002	0.000	0.000	0.000	0.000	0.000
24	FD-916	0.044	1.825	0.004	0.010	0.000	0.001	1.861	0.113	1.136	0.078	0.004	0.000	0.001	0.003	0.001	0.000
25	FD-920	0.010	0.665	0.000	0.006	0.000	0.000	0.433	0.113	0.210	0.078	0.000	0.000	0.001	0.000	0.000	0.000
26	FD-921	0.372	0.938	0.002	0.004	0.000	0.000	0.462	0.113	0.144	0.078	0.001	0.000	0.002	0.000	0.000	0.000
27	FD-926	0.007	0.851	0.000	0.001	0.000	0.000	1.622	0.113	1.106	0.078	0.003	0.000	0.000	0.001	0.000	0.000
28	FD-945	0.851	0.225	0.005	0.004	0.002	0.002	0.075	0.113	0.063	0.209	0.000	0.000	0.002	0.000	0.000	0.000
29	FD-952	0.006	0.732	0.000	0.000	0.000	0.000	0.338	0.247	0.410	0.078	0.001	0.000	0.000	0.000	0.000	0.000

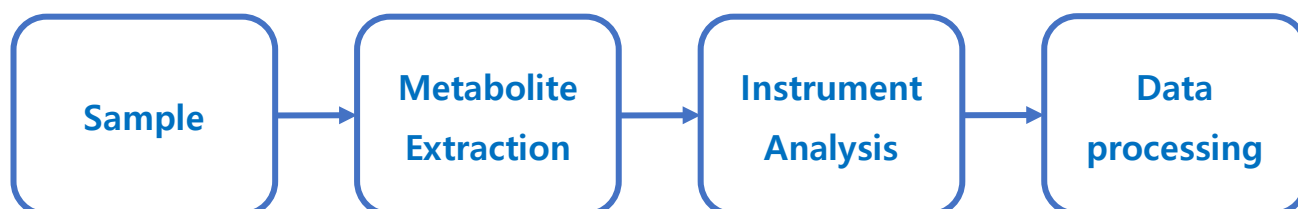
30	FD-1016	0.002	0.469	0.001	0.001	0.000	0.000	0.126	0.113	0.063	0.078	0.006	0.000	0.001	0.000	0.000	0.000
31	FD-1033	0.000	0.057	0.000	0.000	0.000	0.000	0.064	0.113	0.218	0.078	0.000	0.000	0.000	0.000	0.000	0.000
32	FD-1046	0.005	1.576	0.001	0.002	0.000	0.000	1.692	0.113	2.076	0.078	0.007	0.002	0.000	0.000	0.000	0.000
33	FD-1078	0.022	0.690	0.000	0.002	0.000	0.001	0.864	0.113	0.739	0.078	0.002	0.001	0.000	0.002	0.001	0.000
34	FD-1079	0.006	1.785	0.000	0.000	0.000	0.000	1.696	0.113	1.119	0.078	0.002	0.000	0.000	0.002	0.001	0.000

» **Abbreviations** : CA, Cholic acid; CDCA, Chenodeoxycholic acid; GCA, Glycocholic acid; GCDCA, Glycochenodeoxycholic acid; TCA, Taurocholic acid; TCDCA, Taurochenodeoxycholic acid; DCA, Deoxycholic acid; HDCA, Hyodeoxycholic acid; LCA, Lithocholic acid; UDCA, Ursodeoxycholic acid; GDCA, Glycodeoxycholic acid; GLCA, Glycolithocholic acid; GUDCA, Glycoursodeoxycholic acid; TDCA, Taurodeoxycholic acid; TLCA, Tauroolithocholic acid; TUDCA+THDCA, Tauroursodeoxycholic acid+ Taurohyodeoxycholic acid

» **Note:** TUDCA와 THDCA는 구조적 이성질체로서 본 분석 조건에서는 개별 분리가 불가하여, 두 물질을 합산한 농도로 산출함.

» **Concentration (μmol/g)** : 분변 g 당 검출된 각 담즙산의 농도(μmol)

01. Sample preparation and Data analysis



- **Sample** : 시료는 입고 즉시 시료의 변질을 최소화하기 위해 실험 전까지 deep-freezer(-80 °C)에 보관합니다.
- **Metabolite extraction** : 분변 50 mg에 50% 메탄올을 첨가하여 충분히 볼텍싱 후 다음 원심분리를 합니다. 상층액 필터 후, 96 well plate에 옮깁니다
- **Instrument analysis** : Instrument information 에 따라 LC-MS/MS 분석을 수행합니다.
- **Data processing** : 분변에서 검출된 대사체에 대한 calibration curve를 구축하고, 상관계수 $r > 0.999$ 의 값을 확인하며 해당 curve를 기반으로 분변 내 각 대사체에 대한 정량 값을 산출합니다.

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